Volume III · Chapter 14 — Training Pipeline: Pre-training → Fine-tuning → Alignment (DPO) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-14-context.md` for the AI interaction log.

Prerequisites: Ch 11, Ch 12.

Learning outcomes — the student can: describe the full three-stage pipeline; reason about data/compute for pretraining; run domain fine-tuning; build preference data and apply DPO; compare RLHF vs. DPO; manage training runs with tracking.

Topics (theory) — 16 h:

#	Topic	h
14.1	The three-stage pipeline overview (pretrain→SFT→align)	1
14.2	Pretraining: data scale, objectives, tokenization, compute	2
14.3	Domain fine-tuning on medical corpora (e.g., MedFineWeb)	2
14.4	Instruction/supervised fine-tuning data design	2
14.5	Preference alignment: RLHF overview	2
14.6	Direct Preference	3

#	Topic	h
	Optimization (DPO): theory & practice	
14.7	Safety alignment for medical outputs	2
14.8	Training infra: checkpoints, mixed precision, distributed basics, experiment tracking	2

Datasets/tools: HF `transformers/trl/peft`, DPO; experiment tracking (W&B or MLflow); GPU. **Assessment:** aligned model across stages (**60%**); training report (**20%**); quiz (**20%**). **Key decisions:** pretrain vs. fine-tune vs. align scope; DPO vs. RLHF; data quality vs. quantity; compute budget. **References:** plan §13 → *Fine-tuning, alignment & benchmarks*; *Medical LLMs*; *Clinical datasets*. **Hours:** Theory **16** + Lab **20** = **36**.